

Principal-Angle Stability of Determinant Flatness: Sharp Surrogate Transfer and the Critical J^{-1} Accuracy Barrier

Liang Wang

School of Artificial Intelligence and Automation

Huazhong University of Science and Technology, Wuhan 430074, P.R. China

wangliang.f@gmail.com

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Abstract

Let $a = (a_\omega)_{\omega \in \Omega} \neq 0$ be a complex coefficient on its prescribed finite support, let $S = \#\Omega$, and put

$$Z = \sum_{\omega \in \Omega} a_\omega, \quad D = \sum_{\omega \in \Omega} |a_\omega|^2, \quad \rho(a) = \frac{|Z|}{\sqrt{SD}}.$$

Thus $\rho(a)$ is the cosine of the projective angle between a and the constant vector. If

$$a = \frac{Z}{S} \mathbf{1} + a_\perp,$$

then the exact orthogonal decomposition is

$$D = \frac{|Z|^2}{S} + \|a_\perp\|_2^2, \quad \mathfrak{F}(a) := \frac{|Z|^2}{D} = S\rho(a)^2.$$

We determine the sharp perturbation law for this angle. If $\|a - b\|_2 \leq \delta \|b\|_2$, $0 \leq \delta < 1$, and $r = \rho(b)$, then, for support dimension at least two,

$$L(r, \delta) \leq \rho(a) \leq U(r, \delta),$$

where, with $\alpha = \arccos r$ and $\gamma = \arcsin \delta$,

$$L(r, \delta) = \cos(\min\{\frac{\pi}{2}, \alpha + \gamma\}), \quad U(r, \delta) = \cos((\alpha - \gamma)_+).$$

Both endpoints are attained. In particular,

$$|\rho(a) - \rho(b)| \leq \delta,$$

with optimal constant 1. The frequently obtained estimate

$$\rho(a) \leq \frac{\rho(b) + \delta}{1 - \delta}$$

is valid, but is not sharp; its denominator is an artifact of separately estimating the numerator and norm.

This exact stability law identifies a critical surrogate barrier. To transfer an upper target $\rho \ll J^{-1}$, relative ℓ^2 error $O(J^{-1})$ is sufficient and, uniformly over surrogates, necessary in order. Merely $o(1)$ error is insufficient: there are two-dimensional examples with $\rho(b_J) = 0$, relative error $\delta_J \rightarrow 0$, and $\rho(a_J) = \delta_J \gg J^{-1}$. Uniformly preserving a two-sided $\rho \asymp J^{-1}$ law from the norm hypothesis alone additionally requires a fixed margin between the error constant and the surrogate's lower constant.

Finally, we attach the theorem to the literal fixed- h_0 TPC-32 determinant packet. On its actual support $S_X \ll Q_X$, a same-frame surrogate satisfying

$$\rho(b_X) \leq \frac{X^{o(1)}}{J_X}, \quad \frac{\|A_X - b_X\|_2}{\|b_X\|_2} \leq \frac{X^{o(1)}}{J_X}$$

would certify

$$\mathfrak{F}(A_X) \leq X^{1/400+o(1)}$$

at the critical scales. This is an exact L1 stability certificate, not a construction of such a surrogate and not a new L2 estimate. The determinant-energy ledger $\lambda_D \leq 2\eta_Z$ and the independent physical-loss ledger $\Lambda_{\text{phys}} < 1/400$ are kept separate.

Keywords: principal angle; determinant flatness; zero mode; surrogate approximation; relative ℓ^2 error; fixed shift; stability threshold.

Literal-frame convention. Every physical statement concerns one prescribed nonzero shift h_0 , the already aggregated TPC-32 determinant coefficient, its actual post-bin support, and its original global normalization. Surrogates are compared only after they have been placed on that same literal support. No statement is specialized to $h_0 = 2$.

1 Introduction

The TPC-32 matched shell reduces one high-row fixed-shift problem to the distinguished zero-frequency concentration of a literal post-bin coefficient A_X [3]. Writing

$$Z_X = \sum_n A_X(n), \quad D_X = \sum_n |A_X(n)|^2,$$

the statistic is

$$\mathfrak{F}_X = \frac{|Z_X|^2}{D_X}.$$

TPC-81 and TPC-82 separate the two arithmetic obligations behind this quotient: determinant energy must survive, and the signed zero mode must cancel [4, 5]. TPC-88 then factors the same statistic into post-bin magnitude spreading and relative ℓ^1 phase cancellation [7].

The present paper asks a different, operational question:

If the literal coefficient is replaced by an analytically or numerically tractable surrogate, how accurately must that surrogate approximate the coefficient before its zero-mode geometry transfers?

There is a canonical Hilbert-space formulation. On the actual support Ω , let

$$e = \frac{\mathbf{1}}{\sqrt{\#\Omega}}.$$

For a nonzero coefficient a , define

$$\rho(a) = \frac{|\langle a, e \rangle|}{\|a\|_2}.$$

This is the cosine of the projective angle from a to the constant line. Since

$$\langle a, e \rangle = \frac{1}{\sqrt{\#\Omega}} \sum_{\omega \in \Omega} a_\omega,$$

one has the exact identity

$$\mathfrak{F}(a) = \#\Omega \rho(a)^2.$$

Thus the zero-frequency gate is a principal-angle gate.

The principal angle is especially well suited to surrogate analysis because it is invariant under nonzero scalar renormalization. More importantly, its perturbation law is exact. If a lies in the relative ball

$$\|a - b\|_2 \leq \delta \|b\|_2, \quad \delta < 1,$$

then the projective angle between a and b is at most $\arcsin \delta$. This yields sharp upper and lower envelopes for $\rho(a)$, and in particular

$$|\rho(a) - \rho(b)| \leq \delta.$$

The constant is best possible.

This elementary fact has a nontrivial consequence for the TPC program. The critical zero-mode scale is $1/J_X$, not merely a quantity tending to zero. Therefore a surrogate with relative ℓ^2 error $o(1)$ may preserve its energy scale while completely failing to preserve the required zero-mode scale. Uniform transfer requires $O(1/J_X)$ accuracy. A two-dimensional tangent construction shows that this order cannot be improved.

1.1 Main contributions

The paper establishes the following.

- (i) It identifies determinant flatness with the squared principal-angle cosine on the prescribed actual support:

$$D = \frac{|Z|^2}{S} + D_\perp, \quad \mathfrak{F} = S\rho^2.$$

- (ii) It gives the sharp two-sided perturbation envelope for ρ under relative ℓ^2 error, including all boundary cases and equality constructions.
- (iii) It audits the simpler quotient estimate

$$\rho(a) \leq \frac{\rho(b) + \delta}{1 - \delta}$$

and shows that the exact projective result is stronger.

- (iv) It proves sharp transfer inequalities for determinant flatness, coherent energy, orthogonal energy, and the zero mode.
- (v) It proves that $O(1/J)$ relative accuracy is the correct uniform threshold for transferring $\rho \ll 1/J$, whereas $o(1)$ is not enough.
- (vi) It shows that preserving a two-sided $\rho \asymp 1/J$ statement needs a constant margin; an error of the same size as the lower signal can erase that signal exactly.
- (vii) It gives a literal same-support L1 certificate for the fixed- h_0 TPC-32 coefficient.
- (viii) It isolates zero padding as a normalization trap: ρ changes under padding although $\mathfrak{F}$ does not.
- (ix) It keeps the determinant-energy and physical-loss ledgers independent.

1.2 Audit levels

We use the established terminology. An L0 result is finite Hilbert-space algebra or an abstract finite counterexample. An L1 result places that algebra on the literal fixed- h_0 coefficient with its actual bins and original normalization. An L2 result is a new growing- X arithmetic estimate for that coefficient. All angle and perturbation theorems below are L0. The same-frame implication for A_X is L1. No surrogate satisfying its hypotheses is constructed, and no L2 estimate is claimed.

2 Exact principal-angle geometry

Let Ω be a nonempty finite set, put $S = \#\Omega$, and equip $\mathbb{C}^\Omega$ with the inner product

$$\langle x, y \rangle = \sum_{\omega \in \Omega} x_\omega \overline{y_\omega},$$

linear in the first variable. Let

$$e = \frac{\mathbf{1}}{\sqrt{S}}, \quad P_0 x = \langle x, e \rangle e, \quad P_\perp = I - P_0.$$

The line $\mathbb{C}e$ is the constant mode on the prescribed frame.

Definition 2.1 (Literal finite statistics). For $a \in \mathbb{C}^\Omega$, define

$$Z(a) = \sum_{\omega \in \Omega} a_\omega, \quad D(a) = \|a\|_2^2, \quad D_\perp(a) = \|P_\perp a\|_2^2.$$

For $a \neq 0$, define

$$\rho(a) = \frac{|Z(a)|}{\sqrt{SD(a)}} = \frac{|\langle a, e \rangle|}{\|a\|_2}, \quad \mathfrak{F}(a) = \frac{|Z(a)|^2}{D(a)}. \quad (1)$$

The zero vector is treated as a separate branch; no quotient by $D(0)$ is used.

Theorem 2.2 (Orthogonal zero-mode decomposition). *For every $a \in \mathbb{C}^\Omega$,*

$$a = \frac{Z(a)}{S} \mathbf{1} + P_\perp a \quad (2)$$

and

$$D(a) = \frac{|Z(a)|^2}{S} + D_\perp(a). \quad (3)$$

If $a \neq 0$, then

$$\rho(a)^2 = \frac{|Z(a)|^2/S}{D(a)} = 1 - \frac{D_\perp(a)}{D(a)}, \quad \mathfrak{F}(a) = S\rho(a)^2. \quad (4)$$

Proof. Since

$$\langle a, e \rangle = \frac{Z(a)}{\sqrt{S}},$$

one has

$$P_0 a = \frac{Z(a)}{\sqrt{S}} \frac{\mathbf{1}}{\sqrt{S}} = \frac{Z(a)}{S} \mathbf{1}.$$

The decomposition (2) is orthogonal. Pythagoras gives (3), and division by $D(a) > 0$ gives (4). $\square$

Corollary 2.3 (Equality cases). *For every nonzero a ,*

$$0 \leq \rho(a) \leq 1, \quad 0 \leq \mathfrak{F}(a) \leq S.$$

Moreover,

$$\rho(a) = 0 \iff Z(a) = 0,$$

and

$$\rho(a) = 1 \iff a \in \mathbb{C}\mathbf{1} \setminus \{0\}.$$

Proof. These statements follow from (4). The equality $\rho(a) = 1$ is equivalent to $P_\perp a = 0$. $\square$

Remark 2.4 (Scale and phase invariance). For every $c \in \mathbb{C} \setminus \{0\}$,

$$\rho(ca) = \rho(a), \quad \mathfrak{F}(ca) = \mathfrak{F}(a).$$

Thus ρ is naturally a function on complex projective space, not on a chosen normalization of the coefficient.

2.1 Relation to the two-level factorization

For comparison with TPC-88, define, for $a \neq 0$,

$$M(a) = \|a\|_1, \quad \mathfrak{c}(a) = \frac{|Z(a)|}{M(a)}, \quad S_{\text{eff}}(a) = \frac{M(a)^2}{D(a)}.$$

Proposition 2.5 (Exact dictionary). *For every nonzero $a \in \mathbb{C}^\Omega$,*

$$\boxed{\rho(a) = \mathfrak{c}(a) \sqrt{\frac{S_{\text{eff}}(a)}{S}}, \quad \mathfrak{F}(a) = \mathfrak{c}(a)^2 S_{\text{eff}}(a) = S \rho(a)^2.} \quad (5)$$

In particular, $\rho(a) \leq \mathfrak{c}(a)$.

Proof. The first identity follows by inserting $M(a)/\sqrt{D(a)} = \sqrt{S_{\text{eff}}(a)}$ into (1). The remaining identities follow by squaring. Since $S_{\text{eff}}(a) \leq S$ by Cauchy–Schwarz, $\rho(a) \leq \mathfrak{c}(a)$. $\square$

Remark 2.6 (Different normalization questions). The coefficient $\mathfrak{c}$ measures signed cancellation relative to the actual post-bin ℓ^1 mass. The coefficient ρ measures the angle to the constant line relative to ℓ^2 energy and actual support. They are equivalent up to subpower factors only when $S_{\text{eff}}/S = X^{o(1)}$; they should not be interchanged without that input.

2.2 The prescribed-frame rule

Proposition 2.7 (Zero-padding law). *Let $a \in \mathbb{C}^\Omega \setminus \{0\}$, and extend a by zero to a larger frame $\tilde{\Omega} \supset \Omega$ of size $\tilde{S} = S + K$. Denote the padded vector by $\tilde{a}$. Then*

$$Z(\tilde{a}) = Z(a), \quad D(\tilde{a}) = D(a), \quad \mathfrak{F}(\tilde{a}) = \mathfrak{F}(a), \quad (6)$$

but

$$\boxed{\rho_{\tilde{\Omega}}(\tilde{a}) = \sqrt{\frac{S}{S+K}} \rho_{\Omega}(a).} \quad (7)$$

Proof. Padding changes neither the sum nor the squared norm, proving (6). Apply $\mathfrak{F} = S\rho^2$ on the two frames to obtain (7). $\square$

Remark 2.8 (Why actual support is mandatory). Arbitrary padding can make ρ as small as desired without changing the physical coefficient or its flatness. Therefore a claim such as $\rho \ll 1/J$ has content only after the frame has been prescribed. Throughout the physical crosswalk, that frame is the actual support of the literal determinant coefficient. The present theorem applies to an external-grid surrogate only after an exact identification has produced a vector on this literal frame. External coordinates must otherwise be handled by a separately stated union-frame theorem; no such cross-frame transfer is claimed here.

3 Sharp relative- ℓ^2 perturbation theory

For nonzero vectors x, y in a complex Hilbert space, define their projective angle by

$$\angle(x, y) = \arccos \frac{|\langle x, y \rangle|}{\|x\|_2 \|y\|_2} \in [0, \pi/2].$$

This angle is unchanged by multiplying either vector by a nonzero complex scalar. It is the standard one-dimensional principal angle and the Fubini–Study distance between the corresponding projective points; see, for example, Bhatia [1], Stewart and Sun [2].

Lemma 3.1 (A relative ball is a projective cone). *Let $b \neq 0$ and suppose*

$$\|a - b\|_2 \leq \delta \|b\|_2, \quad 0 \leq \delta < 1.$$

Then $a \neq 0$ and

$$\boxed{\angle(a, b) \leq \arcsin \delta.} \quad (8)$$

Proof. The reverse triangle inequality gives $\|a\|_2 \geq (1 - \delta)\|b\|_2 > 0$. Let $\hat{b} = b/\|b\|_2$. Since $a/\|b\|_2$ belongs to the complex line $\mathbb{C}a$,

$$\text{dist}(\hat{b}, \mathbb{C}a) \leq \left\| \hat{b} - \frac{a}{\|b\|_2} \right\|_2 \leq \delta.$$

Orthogonal projection onto $\mathbb{C}a$ gives

$$\text{dist}(\hat{b}, \mathbb{C}a) = \sqrt{1 - \frac{|\langle a, b \rangle|^2}{\|a\|_2^2 \|b\|_2^2}} = \sin \angle(a, b).$$

Because the angle lies in $[0, \pi/2]$, this proves (8). $\square$

Theorem 3.2 (Sharp principal-angle envelope). *Let Ω have size $S \geq 2$, let $a, b \in \mathbb{C}^\Omega$ with $b \neq 0$, and suppose*

$$\|a - b\|_2 \leq \delta \|b\|_2, \quad 0 \leq \delta < 1. \quad (9)$$

Put

$$r = \rho(b), \quad \alpha = \arccos r, \quad \gamma = \arcsin \delta.$$

Then

$$\boxed{L(r, \delta) \leq \rho(a) \leq U(r, \delta),} \quad (10)$$

where

$$\begin{aligned} L(r, \delta) &= \cos \left(\min \left\{ \frac{\pi}{2}, \alpha + \gamma \right\} \right) \\ &= (r\sqrt{1 - \delta^2} - \delta\sqrt{1 - r^2})_+, \end{aligned} \quad (11)$$

$$\begin{aligned} U(r, \delta) &= \cos((\alpha - \gamma)_+) \\ &= \begin{cases} r\sqrt{1 - \delta^2} + \delta\sqrt{1 - r^2}, & \delta < \sqrt{1 - r^2}, \\ 1, & \delta \geq \sqrt{1 - r^2}. \end{cases} \end{aligned} \quad (12)$$

For every $r \in [0, 1]$ and $\delta \in [0, 1)$, both endpoints allowed by (10) are attained by vectors in a two-dimensional subspace. Hence the envelope is sharp.

Proof. Let $e = \mathbf{1}/\sqrt{S}$. Since

$$\rho(x) = \cos \angle(x, e),$$

put $\theta = \angle(a, e)$. The projective angle obeys the triangle inequality. By lemma 3.1,

$$\max\{0, \alpha - \gamma\} \leq \theta \leq \min \left\{ \frac{\pi}{2}, \alpha + \gamma \right\}.$$

The cosine is decreasing on $[0, \pi/2]$, which gives the first forms of L and U . The addition formulas give (11) and the first branch of (12). The lower expression is positive exactly when $\delta < r$; the upper ray reaches the constant line exactly when $\delta \geq \sqrt{1 - r^2}$.

It remains to prove sharpness. Normalize $\|b\|_2 = 1$, multiply by a global phase, and choose a unit vector $f \perp e$ so that

$$b = \cos \alpha e + \sin \alpha f.$$

Such an f exists because $S \geq 2$. When $\sin \alpha = 0$, choose any unit vector orthogonal to e .

If $\alpha \geq \gamma$, define

$$u_+ = \cos(\alpha - \gamma)e + \sin(\alpha - \gamma)f, \quad a_+ = \cos \gamma u_+.$$

Then a_+ is the orthogonal projection of b onto $\mathbb{C}u_+$, so

$$\|a_+ - b\|_2 = \sin \gamma = \delta$$

and $\rho(a_+) = \cos(\alpha - \gamma) = U(r, \delta)$. If $\alpha < \gamma$, the constant ray is within distance $\sin \alpha \leq \delta$ of b ; its projection $a_+ = \cos \alpha e$ attains $\rho(a_+) = 1$.

For the lower endpoint, if $\alpha + \gamma \leq \pi/2$, define

$$u_- = \cos(\alpha + \gamma)e + \sin(\alpha + \gamma)f, \quad a_- = \cos \gamma u_-.$$

Again $\|a_- - b\|_2 = \delta$, and $\rho(a_-) = \cos(\alpha + \gamma) = L(r, \delta)$. If $\alpha + \gamma > \pi/2$, the orthogonal ray $\mathbb{C}f$ is within distance $\cos \alpha = r \leq \delta$ of b ; the projection $a_- = \sin \alpha f$ attains $\rho(a_-) = 0$. The endpoint cases follow by continuity or the same formulas. $\square$

Remark 3.3 (The one-dimensional exception). If $S = 1$, every nonzero vector lies on the constant line and $\rho \equiv 1$. The inequalities above remain valid as bounds, but their lower endpoint need not be attainable. All nontrivial sharpness phenomena begin in dimension two.

Corollary 3.4 (Optimal additive stability). *Under the hypotheses of [theorem 3.2](#),*

$$\boxed{|\rho(a) - \rho(b)| \leq \delta.} \quad (13)$$

The constant 1 is optimal for every $0 < \delta < 1$.

Proof. The angle $\angle(a, e)$ differs from $\angle(b, e)$ by at most $\gamma = \arcsin \delta$. Among two angles in $[0, \pi/2]$ at distance at most γ , the largest possible difference of their cosines is $\sin \gamma = \delta$. Equivalently, this follows directly from (11)–(12).

For sharpness choose orthonormal e, f , put $b = f$, and set

$$u = \delta e + \sqrt{1 - \delta^2} f, \quad a = \sqrt{1 - \delta^2} u.$$

Then

$$\|a - b\|_2 = \delta \|b\|_2, \quad \rho(b) = 0, \quad \rho(a) = \delta.$$

$\square$

3.1 The quotient bound and its exact audit

Corollary 3.5 (Elementary quotient bounds). *Under (9),*

$$\frac{(\rho(b) - \delta)_+}{1 + \delta} \leq \rho(a) \leq \min \left\{ 1, \frac{\rho(b) + \delta}{1 - \delta} \right\}. \quad (14)$$

In particular,

$$\boxed{\rho(a) \leq \frac{\rho(b) + \delta}{1 - \delta}.} \quad (15)$$

These bounds are valid but generally not sharp. The stronger consequence of the exact envelope is

$$(\rho(b) - \delta)_+ \leq \rho(a) \leq \min\{1, \rho(b) + \delta\}. \quad (16)$$

Proof. Cauchy–Schwarz gives

$$|Z(a) - Z(b)| = |\langle a - b, \mathbf{1} \rangle| \leq \sqrt{S} \|a - b\|_2 \leq \sqrt{S} \delta \|b\|_2.$$

Also,

$$(1 - \delta) \|b\|_2 \leq \|a\|_2 \leq (1 + \delta) \|b\|_2.$$

Combining the upper numerator estimate with the lower norm estimate, and the lower numerator estimate with the upper norm estimate, proves (14). Equation (16) is [corollary 3.4](#). $\square$

Remark 3.6 (Where $1 - \delta$ came from). The denominator in (15) is produced by optimizing the numerator and denominator independently. Those two triangle inequalities cannot generally be saturated in the same direction. The projective-cone argument retains their joint geometry and removes the artificial denominator.

3.2 Energy, zero mode, and flatness transfer

Proposition 3.7 (Complete elementary transfer ledger). *Under (9),*

$$(1 - \delta)^2 D(b) \leq D(a) \leq (1 + \delta)^2 D(b), \quad (17)$$

$$|Z(a) - Z(b)| \leq \sqrt{S} \delta \sqrt{D(b)}, \quad (18)$$

$$\left| \sqrt{D_\perp(a)} - \sqrt{D_\perp(b)} \right| \leq \delta \sqrt{D(b)}. \quad (19)$$

Moreover,

$$SL(\rho(b), \delta)^2 \leq \mathfrak{F}(a) \leq SU(\rho(b), \delta)^2, \quad (20)$$

and hence

$$\boxed{(\sqrt{\mathfrak{F}(b)} - \sqrt{S} \delta)_+^2 \leq \mathfrak{F}(a) \leq \min \left\{ S, (\sqrt{\mathfrak{F}(b)} + \sqrt{S} \delta)^2 \right\}}. \quad (21)$$

Proof. The norm triangle inequality proves (17). Cauchy–Schwarz proves (18). Since $P_\perp$ is a contraction,

$$\|P_\perp(a - b)\|_2 \leq \|a - b\|_2,$$

and the reverse triangle inequality proves (19). Finally use $\mathfrak{F} = S\rho^2$, [theorem 3.2](#), and [corollary 3.4](#). $\square$

Remark 3.8 (Different accuracy scales). Any $\delta = o(1)$ transfers the ℓ^2 -energy scale multiplicatively through (17). It does not necessarily transfer a much smaller angle $\rho \asymp 1/J$. The latter task requires error at the angle’s own scale. This distinction is the source of the critical surrogate barrier in the next section.

4 The critical surrogate-accuracy threshold

The perturbation theorem is most transparent for a general target scale $\tau \rightarrow 0$. The TPC application later sets $\tau = 1/J$.

Theorem 4.1 (One-sided target transfer). *Let a_j, b_j be nonzero vectors on the same frames of dimension at least two, let $\tau_j > 0$ satisfy $\tau_j \rightarrow 0$, and fix constants $C, K \geq 0$. Suppose*

$$\begin{aligned} \|a_j - b_j\|_2 &\leq \delta_j \|b_j\|_2, \\ \rho(b_j) &\leq C\tau_j, \quad \delta_j \leq K\tau_j \end{aligned} \quad (22)$$

implies, for all sufficiently large j ,

$$\boxed{\rho(a_j) \leq (C + K)\tau_j}. \quad (23)$$

If $C + K > 0$, the asymptotic coefficient $C + K$ cannot be uniformly reduced using only (22).

Proof. Apply (16). For sharpness, take

$$r_j = C\tau_j, \quad \delta_j = K\tau_j$$

for all sufficiently large j , and choose the upper equality case from [theorem 3.2](#). Since $r_j, \delta_j \rightarrow 0$, the first branch of (12) applies and gives

$$\frac{\rho(a_j)}{\tau_j} = C\sqrt{1 - K^2\tau_j^2} + K\sqrt{1 - C^2\tau_j^2} \rightarrow C + K.$$

Thus no smaller uniform asymptotic coefficient follows from the two scalar bounds alone. $\square$

Corollary 4.2 (Critical J^{-1} sufficiency). *If*

$$\rho(b_J) = O(J^{-1}), \quad \frac{\|a_J - b_J\|_2}{\|b_J\|_2} = O(J^{-1}),$$

then

$$\rho(a_J) = O(J^{-1}).$$

The same statement holds with $O(J^{-1})$ replaced throughout by $J^{-1}X^{o(1)}$ along a critical TPC scale $J = J_X$.

Proof. Apply [theorem 4.1](#) with $\tau_J = J^{-1}$. A sum of two subpower multiples of J_X^{-1} is again a subpower multiple of J_X^{-1} . $\square$

Theorem 4.3 (Two-sided scale transfer with margin). *Suppose, for fixed $0 < c \leq C$ and $0 \leq \vartheta < 1$,*

$$\frac{c}{J} \leq \rho(b_J) \leq \frac{C}{J}, \quad \delta_J \leq \frac{\vartheta c}{J}. \quad (24)$$

Then

$$\boxed{\frac{(1 - \vartheta)c}{J} \leq \rho(a_J) \leq \frac{C + \vartheta c}{J}}. \quad (25)$$

Thus a controlled $O(J^{-1})$ error preserves $\rho \asymp J^{-1}$. In particular, $\delta_J = o(J^{-1})$ preserves every fixed two-sided $\rho(b_J) \asymp J^{-1}$ law.

Proof. Use

$$\rho(b_J) - \delta_J \leq \rho(a_J) \leq \rho(b_J) + \delta_J$$

from [corollary 3.4](#). $\square$

Remark 4.4 (Upper scale versus two-sided scale). An error K/J with arbitrary fixed K always preserves the upper order $O(1/J)$. To preserve a lower bound as well, the error constant must stay below the surrogate's lower constant. Saying only " $O(1/J)$ " suppresses this necessary margin.

4.1 Sharp countermodels

Example 4.5 (Why $o(1)$ is not enough). *Let e, f be orthonormal in $\mathbb{C}^2$, let $\delta_J \rightarrow 0$, and put*

$$b_J = f, \quad u_J = \delta_J e + \sqrt{1 - \delta_J^2} f, \quad a_J = \sqrt{1 - \delta_J^2} u_J. \quad (26)$$

Then

$$\frac{\|a_J - b_J\|_2}{\|b_J\|_2} = \delta_J, \quad \rho(b_J) = 0, \quad \rho(a_J) = \delta_J. \quad (27)$$

Choosing $\delta_J = J^{-1/2}$ gives a surrogate with relative error $o(1)$ and perfect zero mode, while

$$\rho(a_J) = J^{-1/2} \gg J^{-1}.$$

Thus structure-free $o(1)$ relative approximation does not transfer the critical scale.

Remark 4.6 (Actual two-point support). Take

$$e = 2^{-1/2}(1, 1), \quad f = 2^{-1/2}(1, -1).$$

For every sufficiently small positive δ_J , both coordinates of a_J and b_J are nonzero. Hence the countermodel already lives on one common actual support of size two; it is not a padding artifact.

Example 4.7 (Sharp order J^{-1}). *In [example 4.5](#), choose $\delta_J = K/J$ for a fixed $K > 0$. Then*

$$\rho(b_J) = 0, \quad \rho(a_J) = \frac{K}{J}.$$

Consequently the perturbation contribution enters at exactly the J^{-1} scale. A uniform theorem that tolerates $\delta_J \gg J^{-1}$ cannot conclude $\rho(a_J) = O(J^{-1})$.

Example 4.8 (A lower signal can be erased). Fix $c > 0$, let $r_J = c/J < 1$, and set

$$b_J = r_J e + \sqrt{1 - r_J^2} f, \quad a_J = \sqrt{1 - r_J^2} f.$$

Then

$$\rho(b_J) = \frac{c}{J}, \quad \rho(a_J) = 0, \quad \frac{\|a_J - b_J\|_2}{\|b_J\|_2} = \frac{c}{J}.$$

Thus an error with the same leading constant as the lower signal can destroy two-sided comparability. The margin in [theorem 4.3](#) is not cosmetic.

Theorem 4.9 (Sharp order-of-accuracy barrier). Consider all same-frame pairs (a_J, b_J) in dimension at least two. Uniform transfer of

$$\rho(b_J) = O(J^{-1}) \implies \rho(a_J) = O(J^{-1})$$

from a relative-error hypothesis

$$\|a_J - b_J\|_2 \leq \delta_J \|b_J\|_2$$

with a prescribed tolerance $\delta_J \geq 0$ holds if $\delta_J = O(J^{-1})$, and fails in general whenever $J\delta_J$ is unbounded. In particular, failure can occur even when $\delta_J = o(1)$.

Proof. Sufficiency is [corollary 4.2](#). If $J\delta_J$ is unbounded, pass to a subsequence on which it tends to infinity and put

$$\varepsilon_J = \min \left\{ \frac{1}{2}, \sqrt{\frac{\delta_J}{J}} \right\}.$$

Eventually $\delta_J > J^{-1}$, so $\varepsilon_J \leq \delta_J$, while

$$J\varepsilon_J = \min \left\{ \frac{J}{2}, \sqrt{J\delta_J} \right\} \longrightarrow \infty.$$

Use the tangent construction of [example 4.5](#) with actual relative error ε_J on that subsequence. Then $\rho(b_J) = 0$, whereas

$$J\rho(a_J) = J\varepsilon_J$$

is unbounded. The remaining indices may be filled by any exact pair $a_J = b_J \perp e$. □

Remark 4.10 (What the barrier does not say). The theorem is a uniform worst-case statement. A special surrogate may transfer with weaker norm accuracy if one separately proves that its error is almost orthogonal to the constant mode, or proves a signed estimate for $Z(a - b)$ stronger than Cauchy–Schwarz. Such information is additional structure, not a consequence of $\delta = o(1)$.

5 Literal fixed-shift crosswalk

Fix one nonzero integer h_0 . At the high- β TPC-32 endpoint, retain the scales

$$Q_X = X^{267/400+o(1)}, \quad J_X = X^{133/400+o(1)}, \quad C_X = \lfloor J_X \rfloor. \quad (28)$$

Let

$$A_X(n) = A_{C_X}(n)$$

be the literal signed normalized-determinant coefficient after all three matched channels and every native contribution in each determinant bin have been aggregated [\[3\]](#). Define

$$\Omega_X = \text{supp } A_X, \quad S_X = \#\Omega_X, \quad Z_X = \sum_{n \in \Omega_X} A_X(n), \quad D_X = \sum_{n \in \Omega_X} |A_X(n)|^2. \quad (29)$$

TPC-32 gives the imported support envelope

$$S_X \ll X^{o(1)} Q_X. \quad (30)$$

When $A_X \neq 0$, view

$$a_X = (A_X(n))_{n \in \Omega_X} \in \mathbb{C}^{\Omega_X}$$

and define

$$\rho_X = \frac{|Z_X|}{\sqrt{S_X D_X}}, \quad \mathfrak{F}_X = \frac{|Z_X|^2}{D_X} = S_X \rho_X^2. \quad (31)$$

Definition 5.1 (Literal same-frame surrogate). A literal same-frame surrogate is a nonzero vector

$$b_X \in \mathbb{C}^{\Omega_X}$$

whose coordinate $b_X(n)$ corresponds to the same normalized determinant bin n as $A_X(n)$. Define

$$\delta_X = \frac{\|a_X - b_X\|_2}{\|b_X\|_2}, \quad \rho(b_X) = \frac{|\sum_{n \in \Omega_X} b_X(n)|}{\sqrt{S_X} \|b_X\|_2}. \quad (32)$$

No separately chosen surrogate support or channel-wise renormalization is permitted in these definitions.

Theorem 5.2 (Literal surrogate stability certificate). Assume $A_X \neq 0$, $S_X \geq 2$, and $\delta_X < 1$. Then

$$|\rho_X - \rho(b_X)| \leq \delta_X. \quad (33)$$

In particular, if

$$\rho(b_X) \leq \frac{X^{o(1)}}{J_X}, \quad \delta_X \leq \frac{X^{o(1)}}{J_X}, \quad (34)$$

then

$$\boxed{\rho_X \leq \frac{X^{o(1)}}{J_X}, \quad \mathfrak{F}_X \leq X^{1/400+o(1)}}. \quad (35)$$

Proof. Equation (33) is [corollary 3.4](#) on the frame Ω_X . The hypotheses (34) therefore give

$$\rho_X \leq \frac{X^{o(1)}}{J_X}.$$

Using (30),

$$\mathfrak{F}_X = S_X \rho_X^2 \leq X^{o(1)} \frac{Q_X}{J_X^2}.$$

The critical exponents in (28) give

$$\frac{Q_X}{J_X^2} = X^{(267-266)/400+o(1)} = X^{1/400+o(1)}.$$

□

Remark 5.3 (Logical status of the certificate). [Theorem 5.2](#) is a deterministic implication. It does not construct b_X , bound δ_X , or prove its zero-mode angle. Verifying (34) for the growing literal fixed-shift family is precisely additional arithmetic or structural work.

Remark 5.4 (A numerical surrogate must resolve the small angle). A computation showing only $\delta_X = o(1)$ can accurately reproduce the determinant energy while missing the critical zero-mode angle by many powers of J_X . To serve as a rigorous certificate without further signed error structure, it must control the global relative ℓ^2 error at $X^{o(1)}/J_X$ resolution on the literal frame.

5.1 Natural-support equivalence

TPC-83 and TPC-88 identify a conditional natural mass–energy branch in which

$$S_X = X^{o(1)} Q_X. \quad (36)$$

Here and below this denotes two-sided comparability up to subpower factors [6, 7].

Proposition 5.5 (Critical angle equivalence on the natural branch). *Assume (36). Then*

$$\boxed{\mathfrak{F}_X \leq X^{1/400+o(1)} \iff \rho_X \leq \frac{X^{o(1)}}{J_X}.} \quad (37)$$

Without the lower half of (36), the angle condition remains sufficient by (30), but the converse need not hold.

Proof. The identity $\mathfrak{F}_X = S_X \rho_X^2$ and (36) give

$$\rho_X^2 = X^{-267/400+o(1)} \mathfrak{F}_X.$$

Thus the flatness exponent $1/400$ is equivalent to

$$\rho_X \leq X^{-133/400+o(1)} = \frac{X^{o(1)}}{J_X}.$$

The one-sided statement follows exactly as in theorem 5.2. $\square$

Proposition 5.6 (Relative- ℓ^1 and principal-angle thresholds). *On the natural post-bin mass branch of TPC-88,*

$$S_{\text{eff}}(A_X) = X^{o(1)} Q_X, \quad S_X = X^{o(1)} Q_X.$$

Consequently,

$$\rho_X = X^{o(1)} \mathfrak{c}_X. \quad (38)$$

Hence the TPC-88 relative-cancellation threshold $\mathfrak{c}_X \leq X^{o(1)}/J_X$ and the present principal-angle threshold are equivalent up to subpower factors on that branch.

Proof. Use the exact dictionary

$$\rho_X = \mathfrak{c}_X \sqrt{\frac{S_{\text{eff}}(A_X)}{S_X}}$$

from proposition 2.5. $\square$

5.2 Direct zero-mode error scale

Proposition 5.7 (Why J^{-1} produces a Q^2 zero-mode error). *Suppose a same-frame surrogate has*

$$S_X \ll X^{o(1)} Q_X, \quad D(b_X) \ll X^{o(1)} Q_X^3 J_X^2,$$

and

$$\delta_X \leq \frac{X^{o(1)}}{J_X}.$$

Then

$$|Z(a_X) - Z(b_X)| \leq X^{o(1)} Q_X^2. \quad (39)$$

Proof. Apply (18):

$$|Z(a_X) - Z(b_X)| \leq \sqrt{S_X} \delta_X \sqrt{D(b_X)} \leq X^{o(1)} Q_X^{1/2} J_X^{-1} (Q_X^3 J_X^2)^{1/2} = X^{o(1)} Q_X^2.$$

$\square$

Remark 5.8. This computation explains the same threshold in the unnormalized zero mode: J_X^{-1} is exactly the relative energy accuracy that converts the natural determinant norm into the Q_X^2 zero-mode scale. The proposition is conditional and provides no such approximation for the physical coefficient.

6 Separate determinant and physical ledgers

The same exponent $1/400$ appears in several interfaces, but the corresponding quantities are not interchangeable. This section records the ledgers explicitly.

6.1 Determinant-energy and zero-mode ledger

On a conditional natural-support branch, write

$$S_X = X^{o(1)} Q_X, \quad (40)$$

and suppose the literal coefficient satisfies

$$|Z_X| \leq X^{-\eta_Z + o(1)} Q_X^2, \quad (41)$$

$$D_X \geq X^{-\lambda_D + o(1)} Q_X^3 J_X^2. \quad (42)$$

These are hypotheses, not results of the present paper.

Proposition 6.1 (Principal-angle exponent dictionary). *Under (40)–(42),*

$$\rho_X \leq \frac{X^{\lambda_D/2 - \eta_Z + o(1)}}{J_X}. \quad (43)$$

Consequently, the recorded bounds certify the critical angle scale $\rho_X \leq X^{o(1)}/J_X$ whenever

$$\boxed{\lambda_D \leq 2\eta_Z}. \quad (44)$$

Proof. Insert (40)–(42) into

$$\rho_X = \frac{|Z_X|}{\sqrt{S_X D_X}}.$$

The denominator has scale

$$\sqrt{S_X D_X} \geq X^{-\lambda_D/2 - o(1)} Q_X^2 J_X.$$

This proves (43). The exponent of X outside J_X^{-1} is nonpositive precisely under (44). $\square$

Remark 6.2 (A sufficient recorded-bounds condition). The condition $\lambda_D \leq 2\eta_Z$ is the TPC-81/TPC-82 compatibility ledger for these particular recorded bounds [4, 5]. It is not claimed to be necessary for the physical packet if either estimate is nonoptimal.

6.2 Surrogate-accuracy ledger

The surrogate parameter

$$\delta_X = \frac{\|a_X - b_X\|_2}{\|b_X\|_2}$$

has two distinct jobs:

- (a) $\delta_X = o(1)$ transfers the determinant-energy scale by (17);
- (b) $\delta_X \leq X^{o(1)}/J_X$ transfers the critical principal-angle scale by (33).

The second is strictly stronger. Neither hypothesis is presently verified for a literal TPC-32 surrogate.

Ledger	Certified comparison	Status here
Determinant angle	$\lambda_D \leq 2\eta_Z$ transfers the recorded Z_X, D_X bounds to $\rho_X \leq X^{o(1)}/J_X$	Conditional L1; no Z_X or D_X estimate proved
Surrogate stability	$\rho(b_X), \delta_X \leq X^{o(1)}/J_X$ transfers the angle and flatness target	Exact L1 implication; no surrogate constructed
Physical reassembly	$\Lambda_{\text{phys}} < 1/400$	Independent inherited stop rule

Table 1: The three comparisons have different inputs and cannot be merged without an explicit crosswalk.

6.3 Physical-loss ledger

The complete physical reduction has its own fixed-power loss Λ_{phys} . The inherited endpoint rule is

$$\boxed{\Lambda_{\text{phys}} < \frac{1}{400}}. \quad (45)$$

Equality or excess is a stop condition.

Proposition 6.3 (No automatic ledger conversion). *The identities and perturbation inequalities of this paper do not imply any of the following conversions:*

- (i) a determinant-energy loss λ_D is a physical reassembly loss;
- (ii) the approximation exponent in δ_X is a saving in η_Z ;
- (iii) a numerical surrogate error may be subtracted from Λ_{phys} ;
- (iv) the inequalities (44) and (45) may be added into one budget.

Proof. The determinant ledger compares the literal post-bin zero mode with literal post-bin energy. The surrogate ledger compares two vectors on that same coefficient space. The physical ledger records losses in the upstream and downstream arithmetic reassembly. No theorem above identifies any pair of these quantities. Therefore no conversion follows. $\square$

Remark 6.4 (What a valid crosswalk would require). To enter a surrogate construction into the physical ledger, one would need a literal theorem that tracks its approximation error through every mask, determinant bin, polarization, and outer normalization. To enter determinant-energy loss into the physical ledger, one would need a theorem expressing that loss as an independent consumption of the final saving. Until such statements are proved, the ledgers remain separate and double counting is forbidden.

7 Claim boundary and next gate

Theorem 7.1 (Audited conclusion). *Let $a \neq 0$ be a finite complex coefficient on its prescribed support of size S . Then*

$$D(a) = \frac{|Z(a)|^2}{S} + D_{\perp}(a), \quad \mathfrak{F}(a) = S\rho(a)^2.$$

If a is approximated by $b \neq 0$ with

$$\|a - b\|_2 \leq \delta \|b\|_2, \quad \delta < 1,$$

then the sharp envelope

$$L(\rho(b), \delta) \leq \rho(a) \leq U(\rho(b), \delta)$$

holds, and in particular

$$|\rho(a) - \rho(b)| \leq \delta.$$

The constant is optimal. Consequently,

$$\rho(b_J) = O(J^{-1}), \quad \delta_J = O(J^{-1}) \implies \rho(a_J) = O(J^{-1}),$$

and this order of relative accuracy is uniformly sharp.

For the literal fixed- h_0 TPC-32 coefficient on its actual support, the conditional same-frame hypotheses

$$\rho(b_X), \delta_X \leq \frac{X^{o(1)}}{J_X}$$

certify

$$\mathfrak{F}_X \leq X^{1/400+o(1)}.$$

Proof. Combine [theorems 2.2, 3.2, 4.9](#) and [5.2](#) and [corollary 3.4](#). □

What is proved. The orthogonal decomposition, exact angle envelope, equality constructions, padding law, J^{-1} accuracy barrier, and finite countermodels are unconditional L0 results. Their placement on the actual fixed- h_0 determinant support is an exact L1 stability certificate.

What is not proved. No same-frame surrogate for A_X is constructed. No bound $\delta_X \leq X^{o(1)}/J_X$ is proved. No determinant-energy lower bound, distinguished zero-mode estimate, growing fixed-shift Möbius-correlation estimate, or full-block remainder estimate is proved. Thus there is no new L2 arithmetic theorem.

Support rule. The angle must be computed on the actual post-bin support of the literal coefficient. Zero padding changes ρ . A surrogate with a different grid or support is covered only after an exact map places it on the literal bins. Otherwise a separate union-frame theorem is required and the present certificate does not apply.

Degenerate branches. If $A_X = 0$, one may set $\mathfrak{F}_X = 0$ by convention, but ρ_X is left undefined and no division by D_X is made. This branch does not certify determinant-energy survival. If $A_X \neq 0$ and $S_X = 1$, then $\rho_X = \mathfrak{F}_X = 1$; the flatness upper target is numerically automatic for large X , while the nontrivial two-dimensional sharpness constructions are inapplicable.

Accuracy rule. Relative error $o(1)$ is sufficient to transfer a nonzero determinant-energy scale, but not a J^{-1} principal-angle scale. For a one-sided upper target, $O(J^{-1})$ is the sharp uniform order. For a two-sided $\asymp J^{-1}$ statement, the error constant must also remain strictly below the lower signal constant.

Ledger rule. The determinant compatibility condition

$$\lambda_D \leq 2\eta_Z$$

and the physical stop rule

$$\Lambda_{\text{phys}} < 1/400$$

belong to different ledgers. The surrogate error belongs to a third stability interface. None may be added to, subtracted from, or substituted for another without a proved literal crosswalk.

Arithmetic claim boundary. Nothing here specializes the prescribed nonzero h_0 to 2, breaches the sieve parity barrier, proves a fixed-shift Hardy–Littlewood asymptotic, gives a prime-pair lower bound, or proves the twin-prime conjecture. The critical-angle certificate only states what an eventual surrogate theorem would have to resolve.

Next gate. The most direct positive continuation is to construct a literal same-frame surrogate b_X for which both

$$\rho(b_X) \leq \frac{X^{o(1)}}{J_X} \quad \text{and} \quad \|A_X - b_X\|_2 \leq \frac{X^{o(1)}}{J_X} \|b_X\|_2$$

can be proved. If only $\delta_X = o(1)$ is accessible, a different route must establish additional signed orthogonality of the error, for example a direct improvement of

$$|Z(A_X - b_X)| \leq \sqrt{S_X} \|A_X - b_X\|_2.$$

Failure of both routes would itself identify a precise nonidentifiability barrier, rather than evidence for a prime-pair claim.

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